PC/CP220 Project Phase III Checklist (3.01)

A. General

1. Professionally presented Neat, etc. ________

2. Properly identified
   (eg. name, id) ________

3. On time
   at *beginning* of lab *with checklist* ________

4. Good grammar
   (eg. complete sentences where required) ________

5. Correct spelling
   ____

B. Content

1. *Circuit diagram is consisted* ________
   with equations

2. *All cases are simulated* ________
   With good explanations

3. *Circuit passes/ fails simulation*
   Noted (matches truth table) ________

4. *All input/ output parts have been*
   Listed (including resistors) ________

5. *All previous phases included*
   (updated if necessary) ________

CP 220 - Project Phase III

Project Topic: Digits of e

Student Name: Moosa Ahmed

Student ID: 169046796

Introduction:

The digit of e Circuit is designed to return the first 10 from 1st to 10th decimal places value of e including the whole value i.e; 2 at 0th decimal value (the integer part). We have taken 4 inputs A0 to A3 and 4 outputs O0 to O3 as defined in the Phase 1 and 2 of the Project.

The digits e = **2.718 281 828 5**.

Truth Table:

This table shows the expected outputs for each input from 0 to 15. (0000 to 1111)

A3	A2	A1	A0	Decimal place	Digits of 'e'	O3	O2	O1	O0
0	0	0	0	0 (integer part)	2	0	0	1	0
0	0	0	1	1	7	0	1	1	1
0	0	1	0	2	1	0	0	0	1
0	0	1	1	3	8	1	0	0	0
0	1	0	0	4	2	0	0	1	0
0	1	0	1	5	8	1	0	0	0
0	1	1	0	6	1	0	0	0	1
0	1	1	1	7	8	1	0	0	0
1	0	0	0	8	2	0	0	1	0
1	0	0	1	9	8	1	0	0	0
1	0	1	0	10	5	0	1	0	1
1	0	1	1	11	Invalid	0	0	0	0
1	1	0	0	12	Invalid	0	0	0	0
1	1	0	1	13	Invalid	0	0	0	0
1	1	1	0	14	Invalid	0	0	0	0
1	1	1	1	15	Invalid	0	0	0	0

Logic Equations:

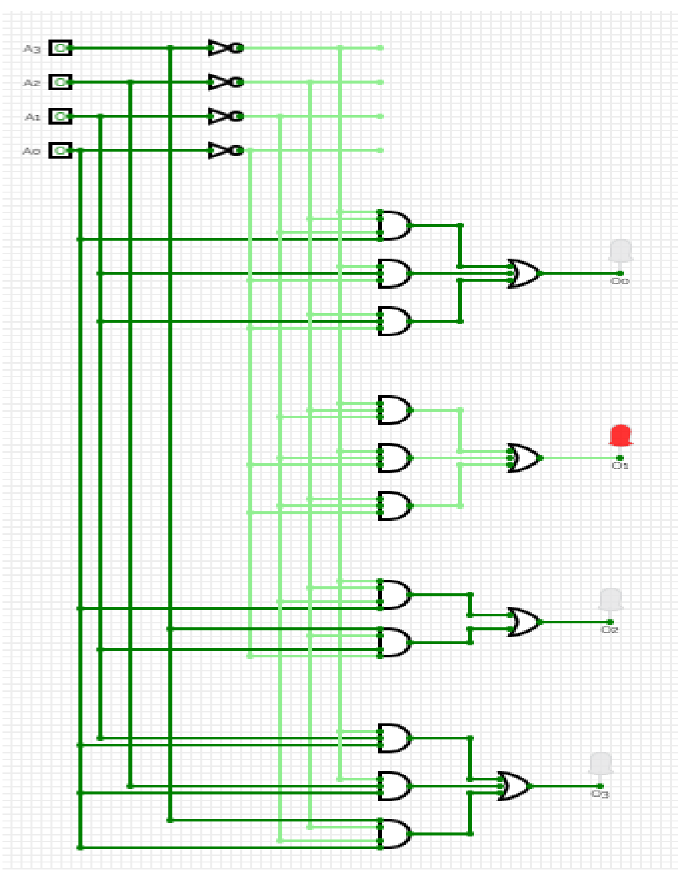
Since we are dealing with a circuit which takes 4 inputs and gives 4 outputs, so we have to have 4 different equations to represent each output.

1. $O0 = A3'A2'A1'A0 + A3'A1A0' + A2'A1A0'$ (minimum SOP)
2. $O1 = A3'A2'A1' + A3'A1'A0' + A2'A1'A0'$ (minimum SOP)
3. $O2 = A3'A2'A1'A0 + A3A2'A1A0'$ (minimum SOP)
4. $O3 = A3'A1A0 + A3'A2A0 + A3A2'A1'A0$. (minimum SOP)

These outputs combined show the binary output.

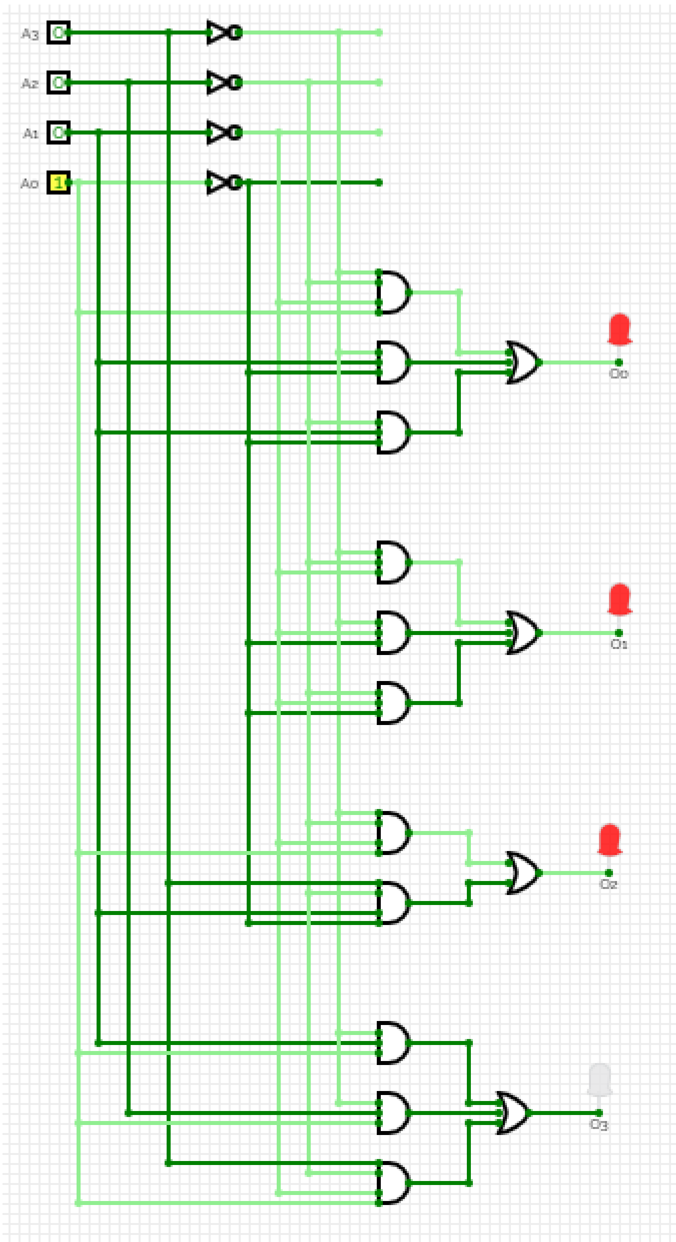
Circuit Diagrams:

input = 0; output = 2:



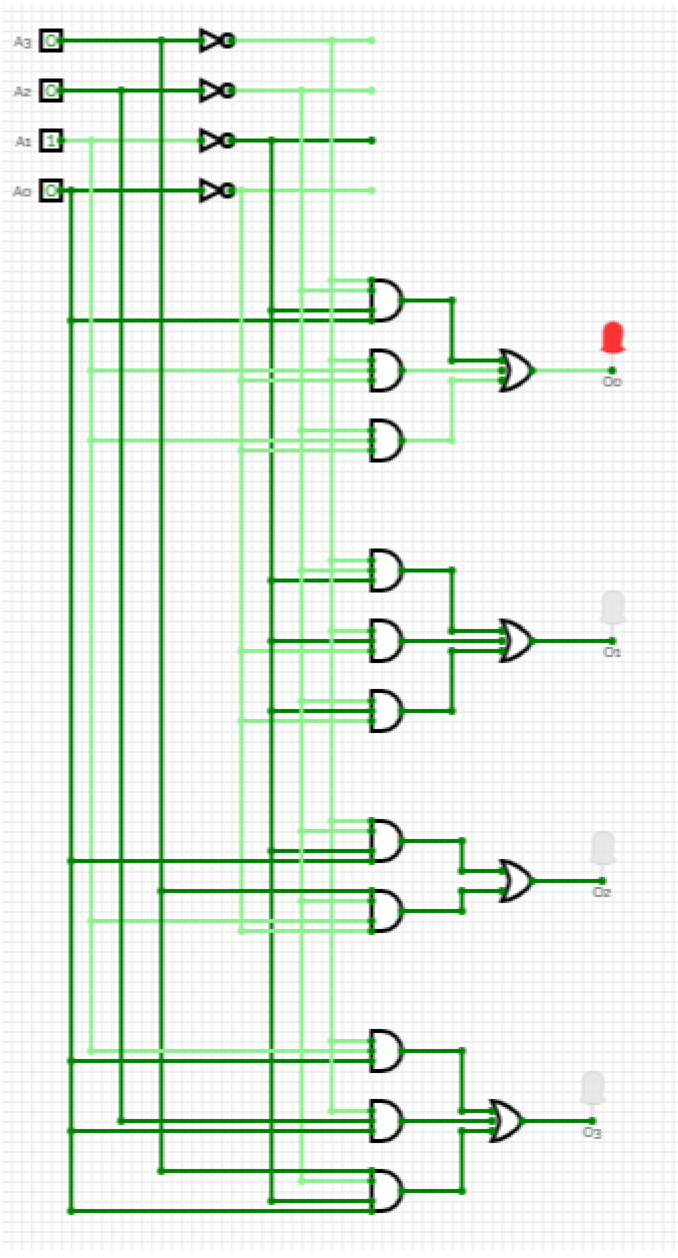
The circuit above matches the required output as show in the truth table. On the 4 bit input of 0000 it outputs 2.

input = 1; output = 7:



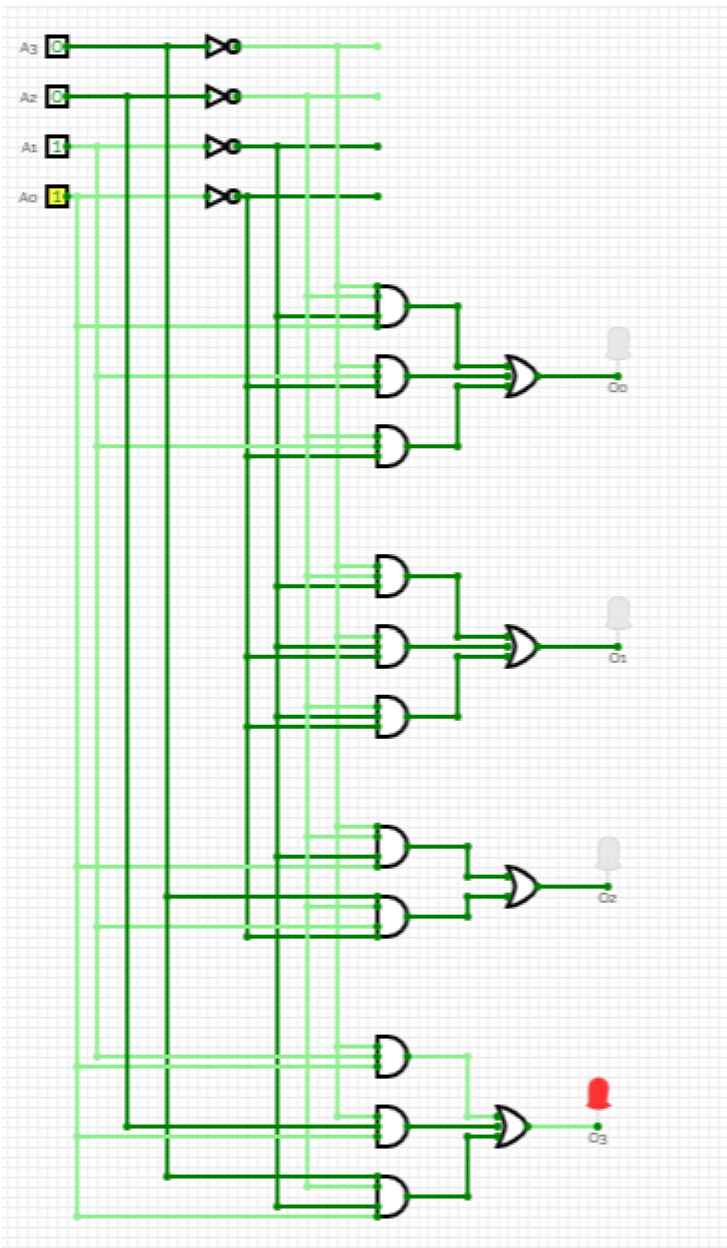
The circuit above matches the required output as show in the truth table. On the 4 bit input of 0001 it outputs 7.

input = 2; output = 1:



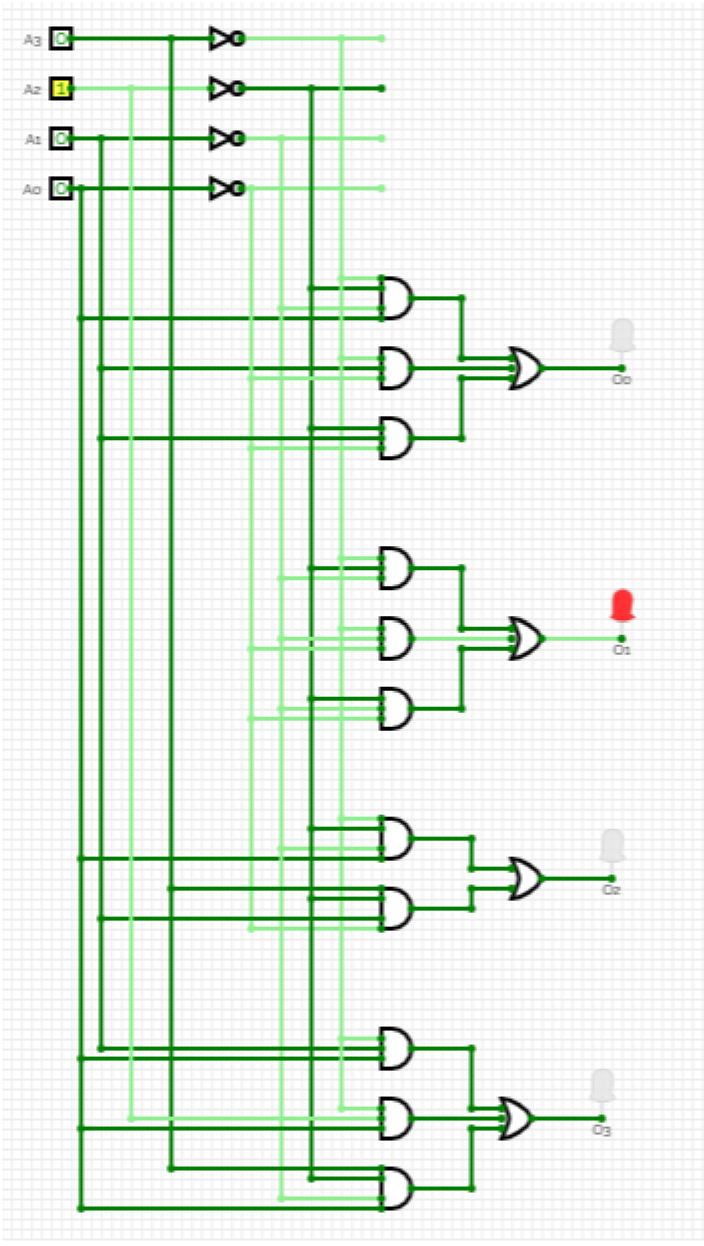
The circuit above matches the required output as show in the truth table. On the 4 bit input of 0010 it outputs 1.

input = 3; output = 8:



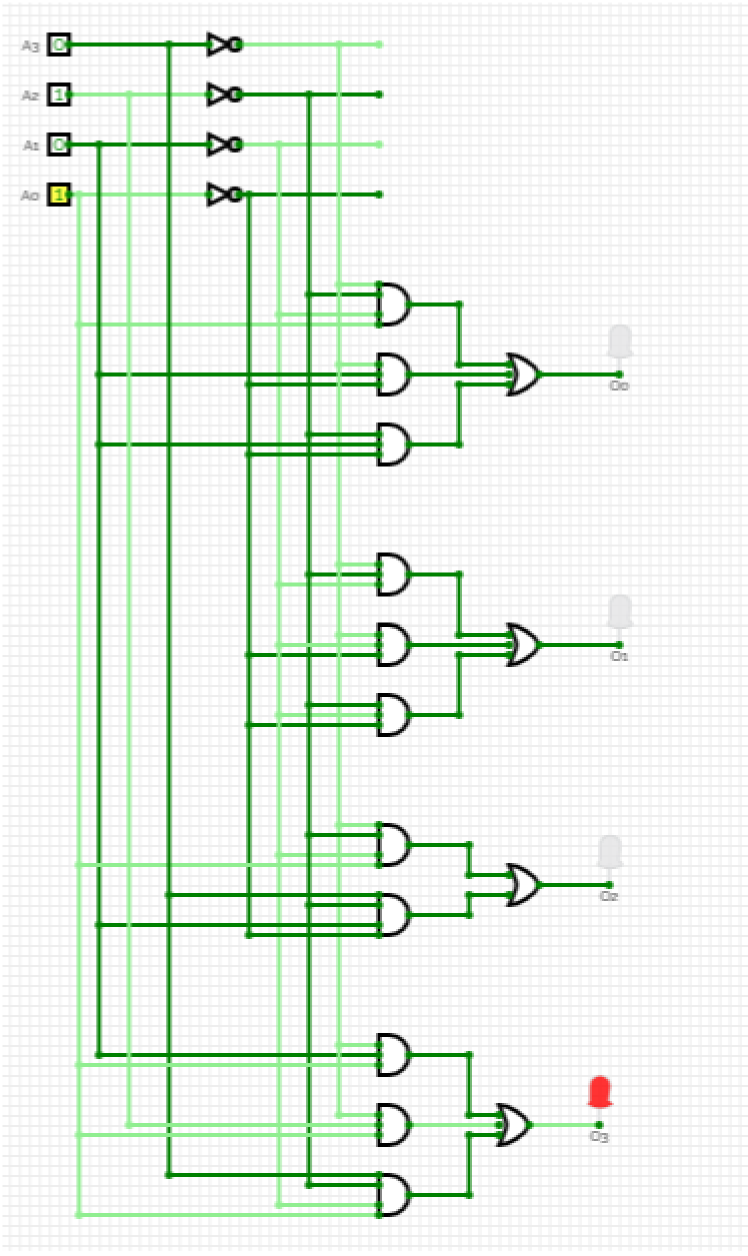
The circuit above matches the required output as show in the truth table. On the 4 bit input of 0011 it outputs 8.

input = 4; output = 2:



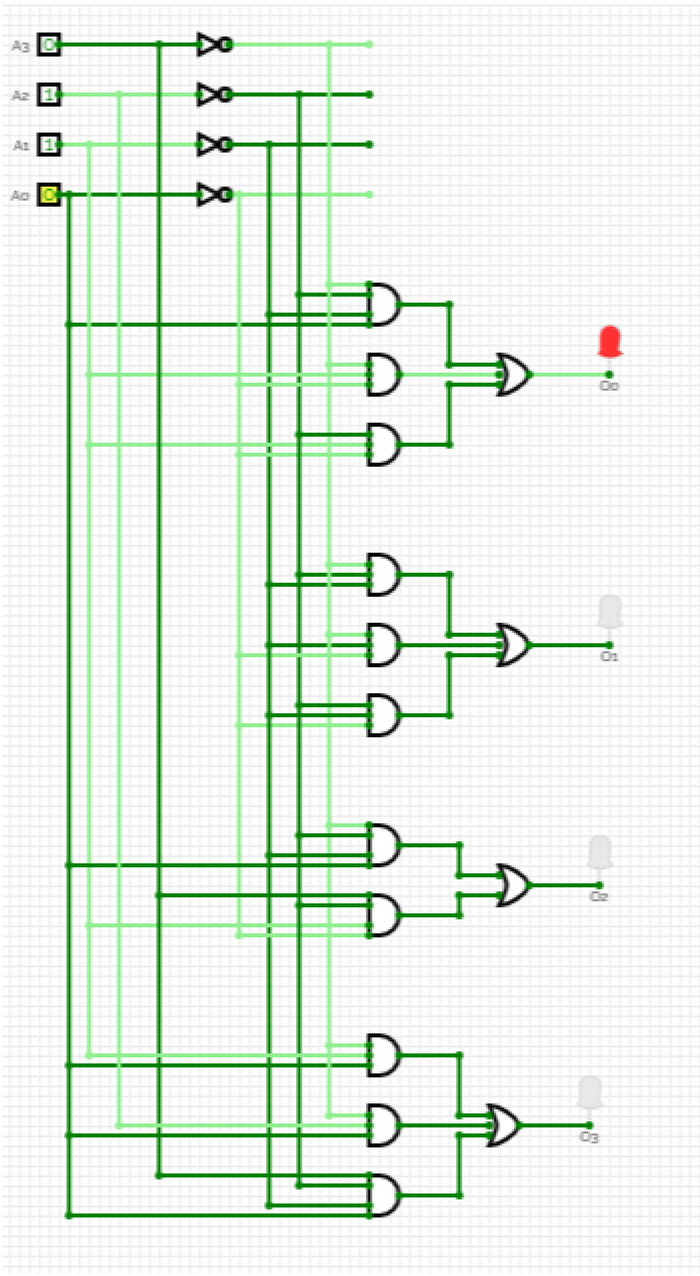
The circuit above matches the required output as show in the truth table. On the 4 bit input of 0100 it outputs 2.

input = 5; output = 8:



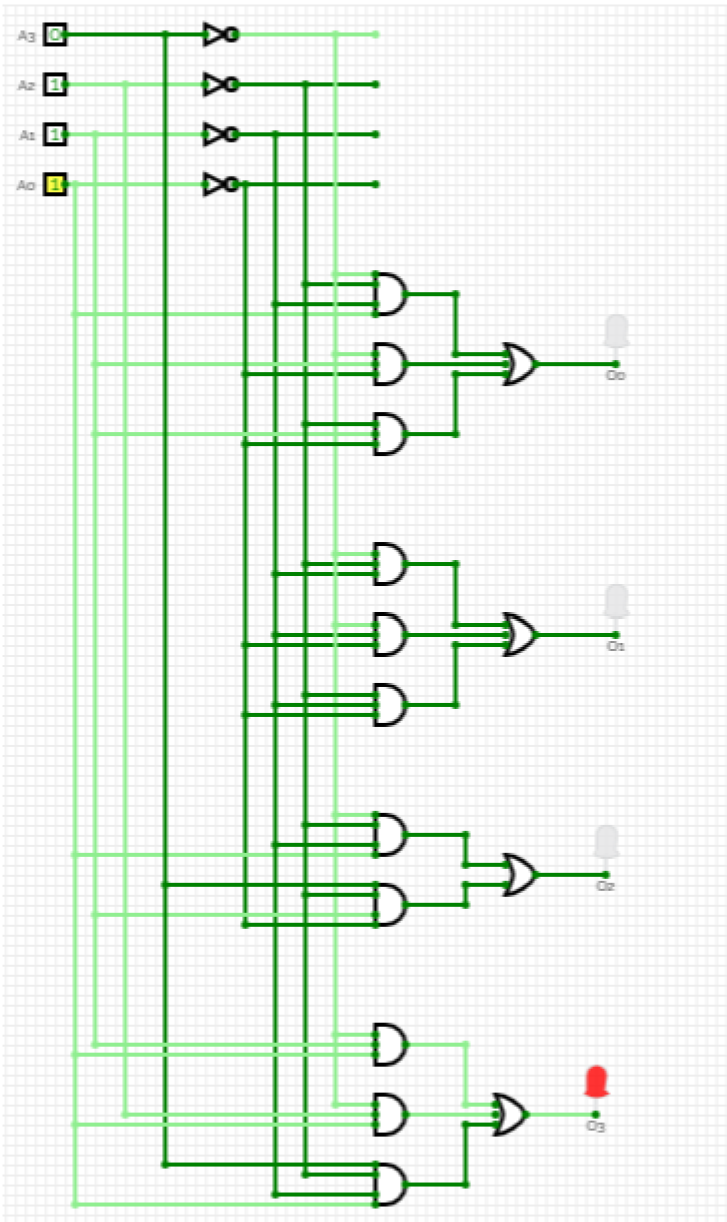
The circuit above matches the required output as show in the truth table. On the 4 bit input of 0101 it outputs 8.

input = 6; output = 1:



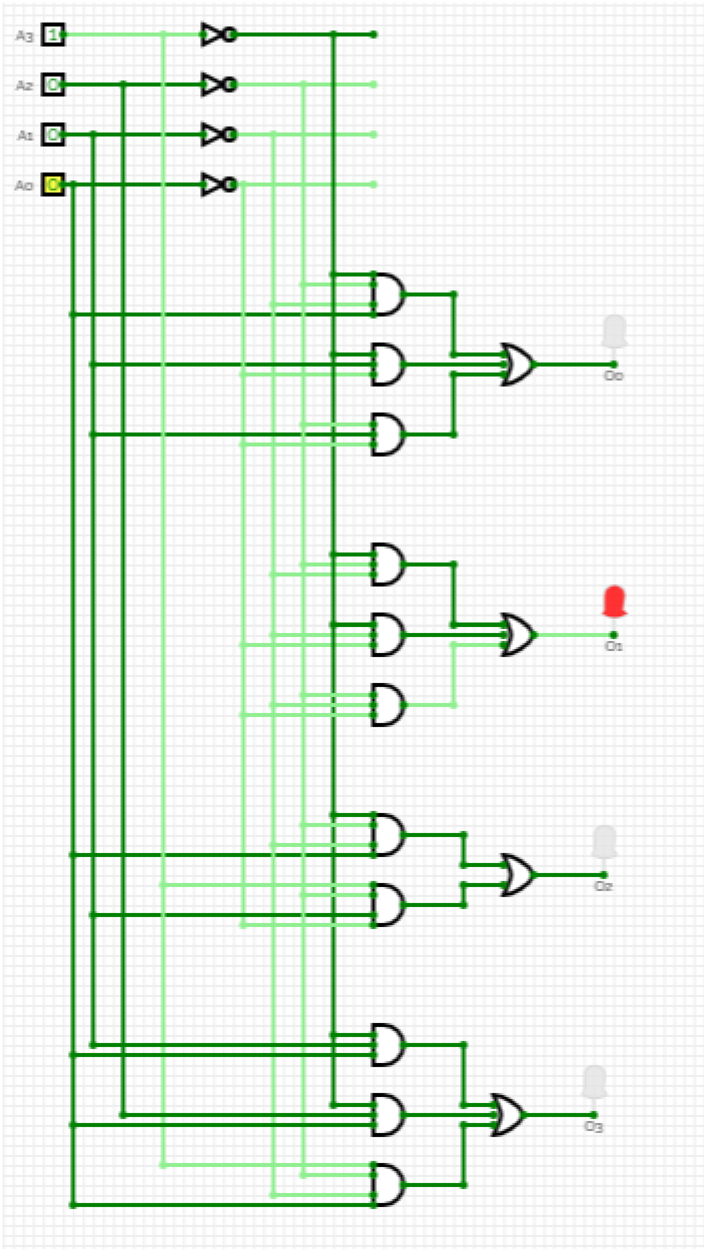
The circuit above matches the required output as show in the truth table. On the 4 bit input of 0110 it outputs 1.

input = 7; output = 8:



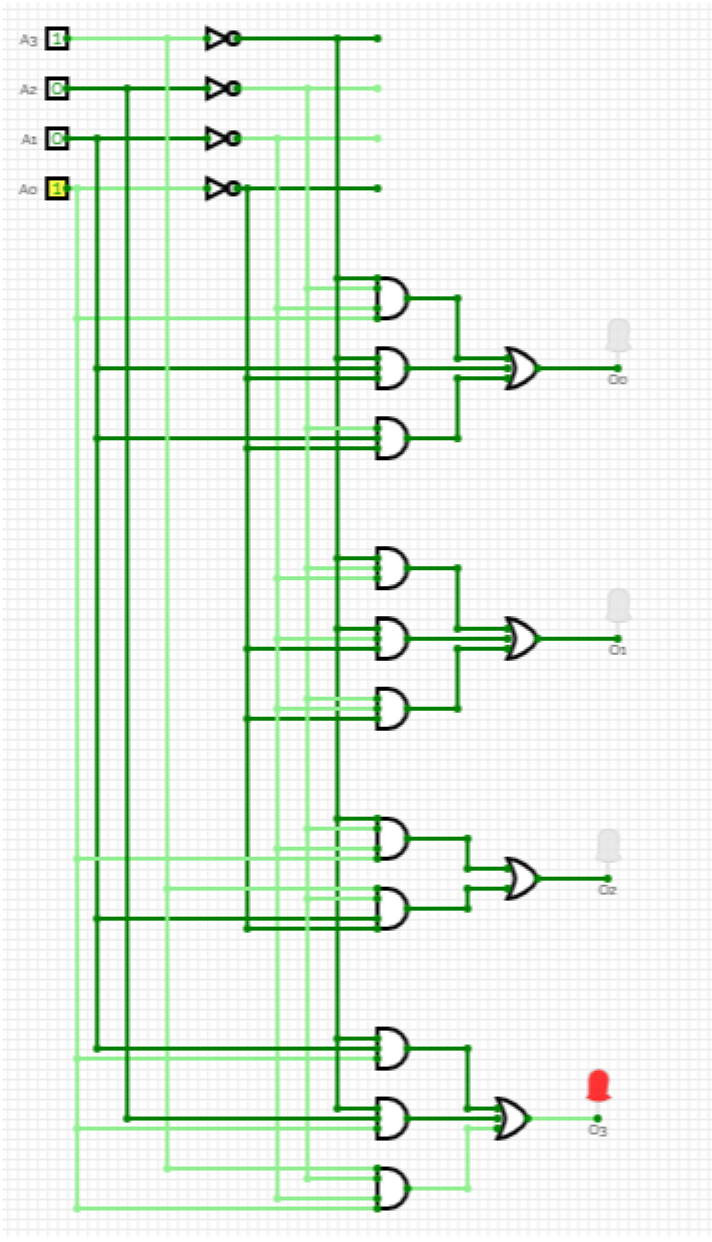
The circuit above matches the required output as show in the truth table. On the 4 bit input of 0111 it outputs 8.

input = 8; output = 2:



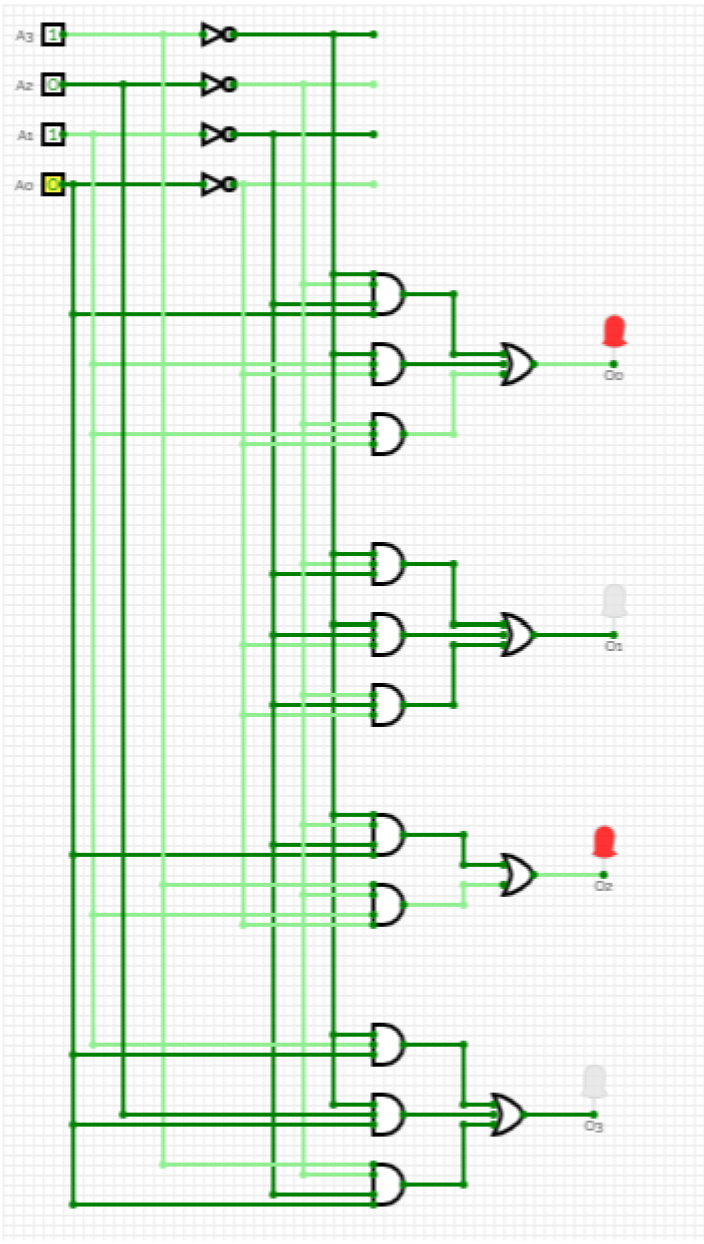
The circuit above matches the required output as show in the truth table. On the 4 bit input of 1000 it outputs 2.

input = 9; output = 8:



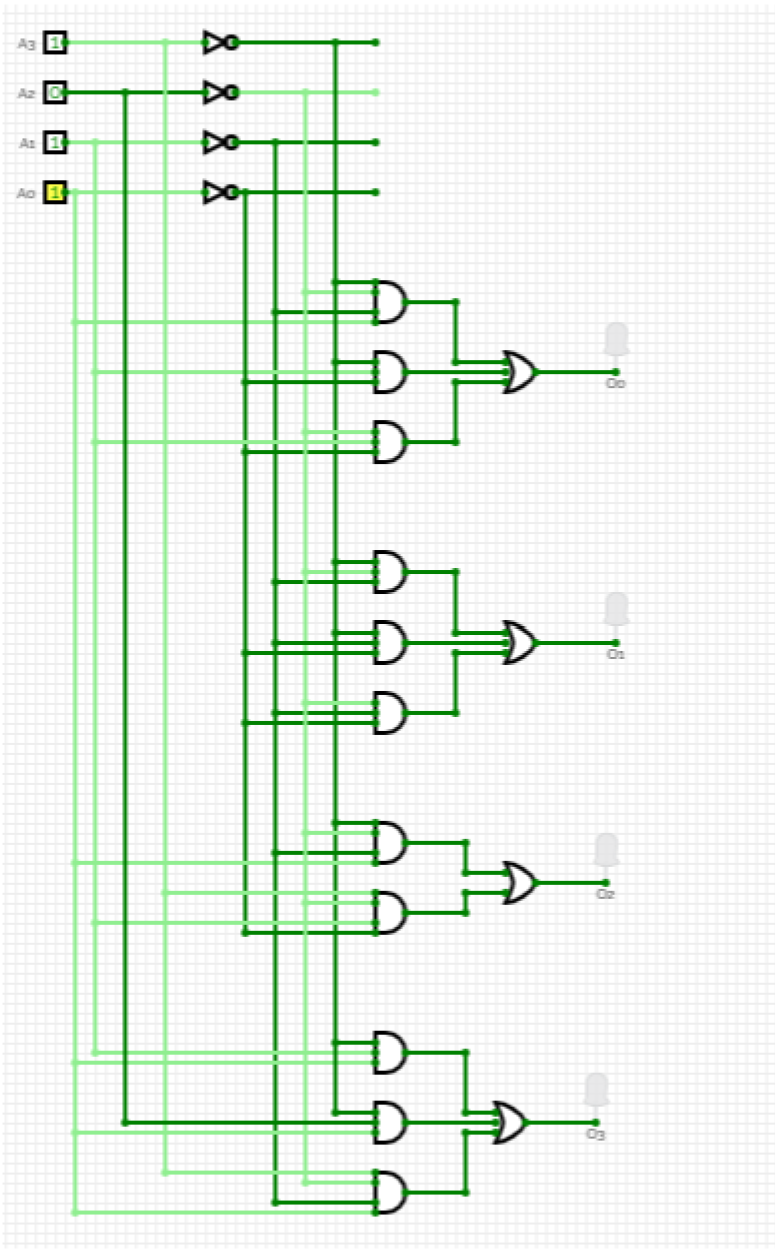
The circuit above matches the required output as show in the truth table. On the 4 bit input of 1001 it outputs 8.

input = 10; output = 5:



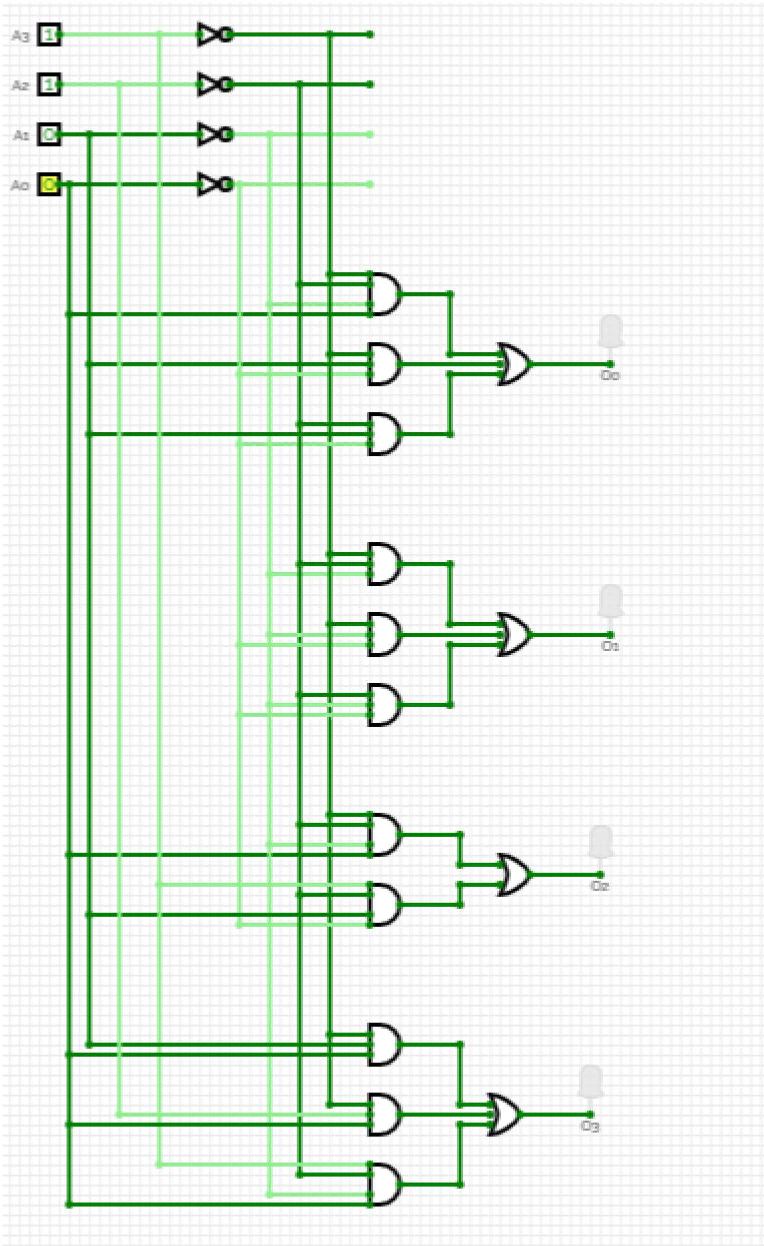
The circuit above matches the required output as show in the truth table. On the 4 bit input of 1010 it outputs 5.

input = 11; output = 0:



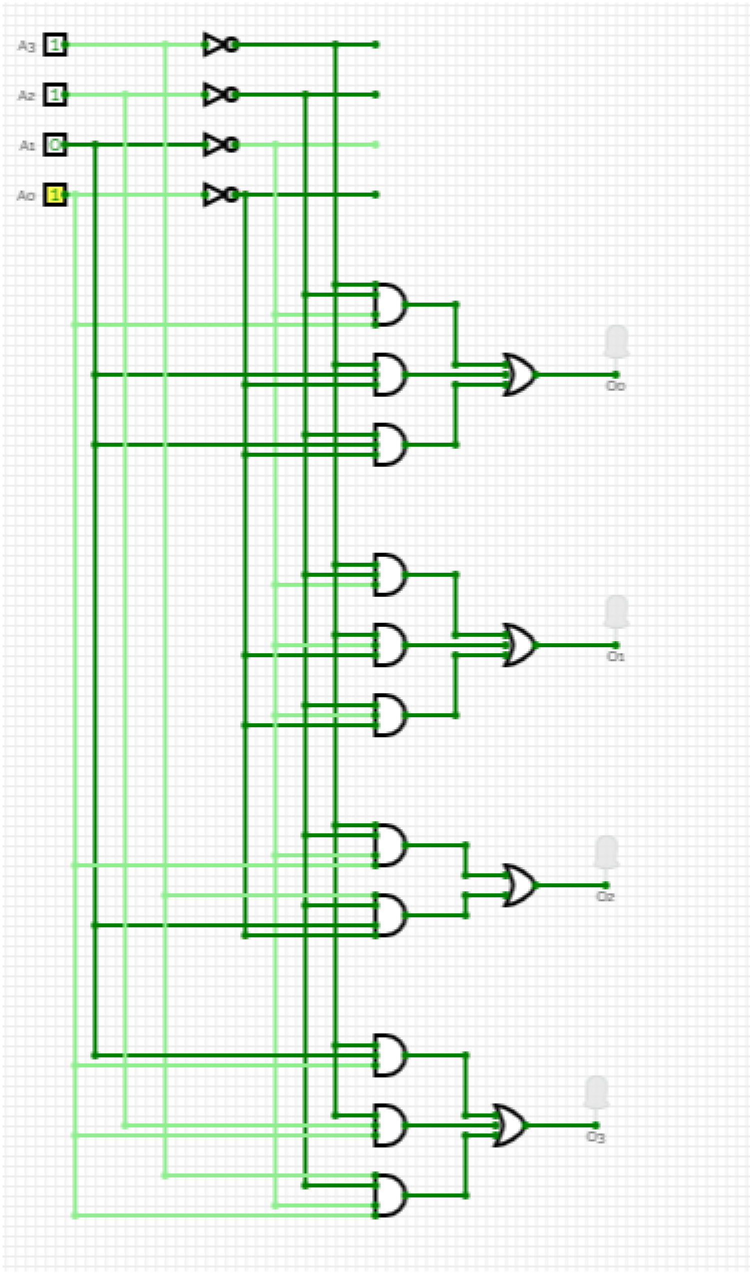
The circuit above matches the required output as show in the truth table. On the 4 bit input of 1011 it outputs 0.

input = 12; output = 0:



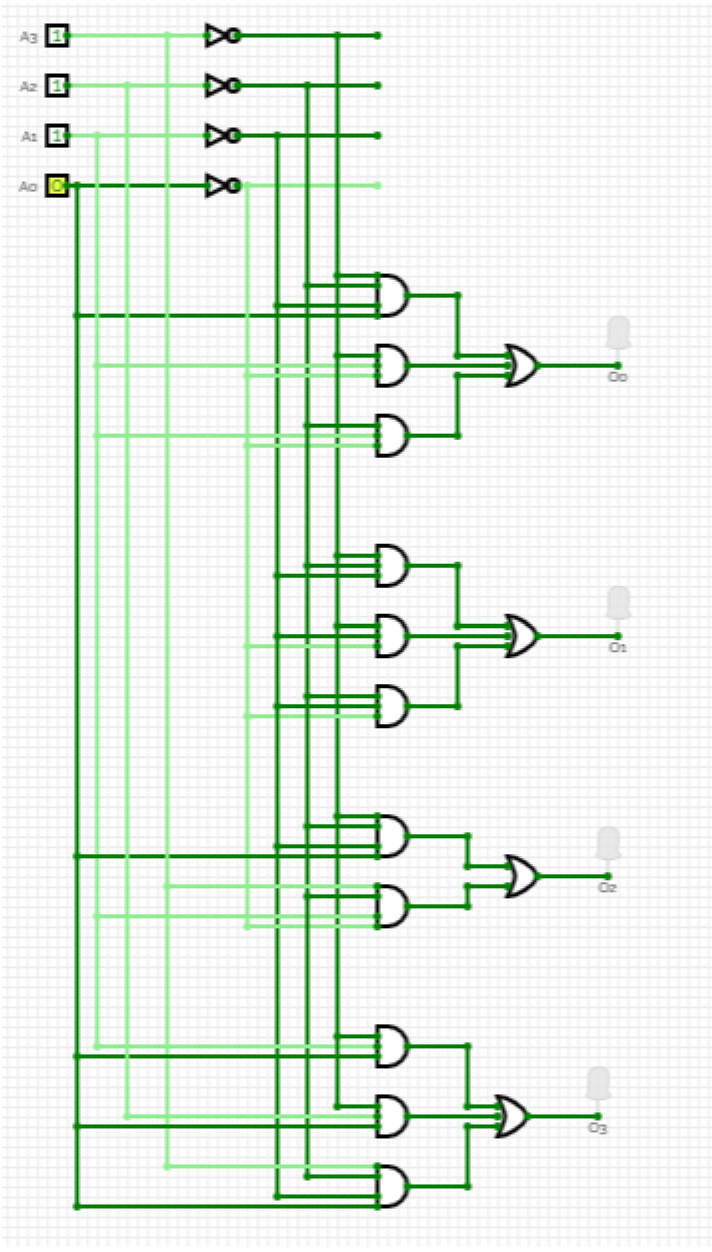
The circuit above matches the required output as show in the truth table. On the 4 bit input of 1100 it outputs 0.

input = 13; output = 0:



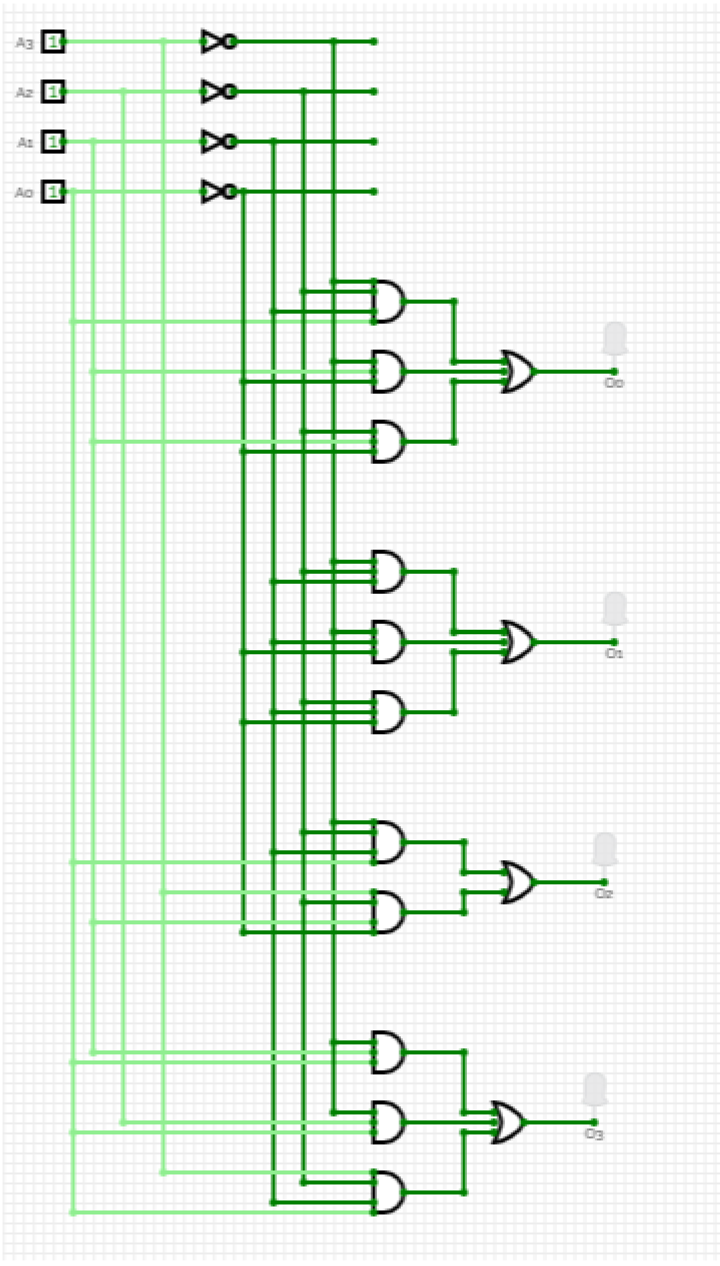
The circuit above matches the required output as show in the truth table. On the 4 bit input of 1101 it outputs 0.

input = 14; output = 0:



The circuit above matches the required output as show in the truth table. On the 4 bit input of 1110 it outputs 0.

Input = 15; output = 0:



The circuit above matches the required output as show in the truth table. On the 4 bit input of 1111 it outputs 0.

Parts List:

- breadboard
- Prototype switches for inputs
- AND, OR, NOT gates
- 4 LEDs for outputs
- 4 resistors (1kΩ)

Conclusion:

The circuit diagrams correctly comply with the expected outputs for inputs from 0 to 15 and matches the output as in the truth table. The first 11 inputs from 0000 (0) to 1010 (10) gives the digits of ‘e’ and the from 1011 (11) to 1111 (15) it shows 0000 (0) as output which indicates the invalid inputs as the circuit was designed to output only first 10 decimal places digits of e including the integer part.

CP 220 - Project Phase II - not revised

Project Topic: Digits of e

Student Name: Moosa Ahmed

Student ID: 169046796

Truth Table:

For this particular problem, we need to have 4 inputs from A0 to A3 and 4 outputs from O0 to O3 as defined in Phase I of the project. It would be helpful to create a table showing 4 inputs and 4 corresponding outputs.

Digits of e: **2.718 281 828 5**.

A3	A2	A1	A0	Decimal place	Digits of 'e'	O3	O2	O1	O0
0	0	0	0	0 (integer part)	2	0	0	1	0
0	0	0	1	1	7	0	1	1	1
0	0	1	0	2	1	0	0	0	1
0	0	1	1	3	8	1	0	0	0
0	1	0	0	4	2	0	0	1	0
0	1	0	1	5	8	1	0	0	0
0	1	1	0	6	1	0	0	0	1
0	1	1	1	7	8	1	0	0	0
1	0	0	0	8	2	0	0	1	0
1	0	0	1	9	8	1	0	0	0
1	0	1	0	10	5	0	1	0	1
1	0	1	1	11	Invalid	0	0	0	0
1	1	0	0	12	Invalid	0	0	0	0
1	1	0	1	13	Invalid	0	0	0	0
1	1	1	0	14	Invalid	0	0	0	0
1	1	1	1	15	Invalid	0	0	0	0

Karnaugh Maps (K - maps):

Since we have 4 different bits for the outputs, O3, O2, O1, O0, we will have to make 4 different K - maps for each output.

K - map for O0: ‘x’ represents the bit we don’t need.

		A1 A0			
		00	01	11	10
A3 A2	00	0	1	0	1
	01	0	0	0	1
	11	0	0	0	0
	10	0	0	0	1

Equation: O0 = A3'A2'A1'A0 + A3'A1A0' + A2'A1A0' (minimum SOP)

K - map for O1: ‘x’ represents the bit we don’t need.

		A1 A0			
		00	01	11	10
A3 A2	00	1	1	0	0
	01	1	0	0	0
	11	0	0	0	0
	10	1	0	0	0

Equation: O1 = A3'A2'A1' + A3'A1'A0' + A2'A1'A0' (minimum SOP)

K - map for O2: ‘x’ represents the bit we don’t need.

		A1 A0			
		00	01	11	10
A3 A2	00	0	1	0	0
	01	0	0	0	0
	11	0	0	0	0
	10	0	0	0	1

Equation: O2 = A3'A2'A1'A0 + A3A2'A1A0’ (minimum SOP)

K - map for O3: ‘x’ represents the bit we don’t need.

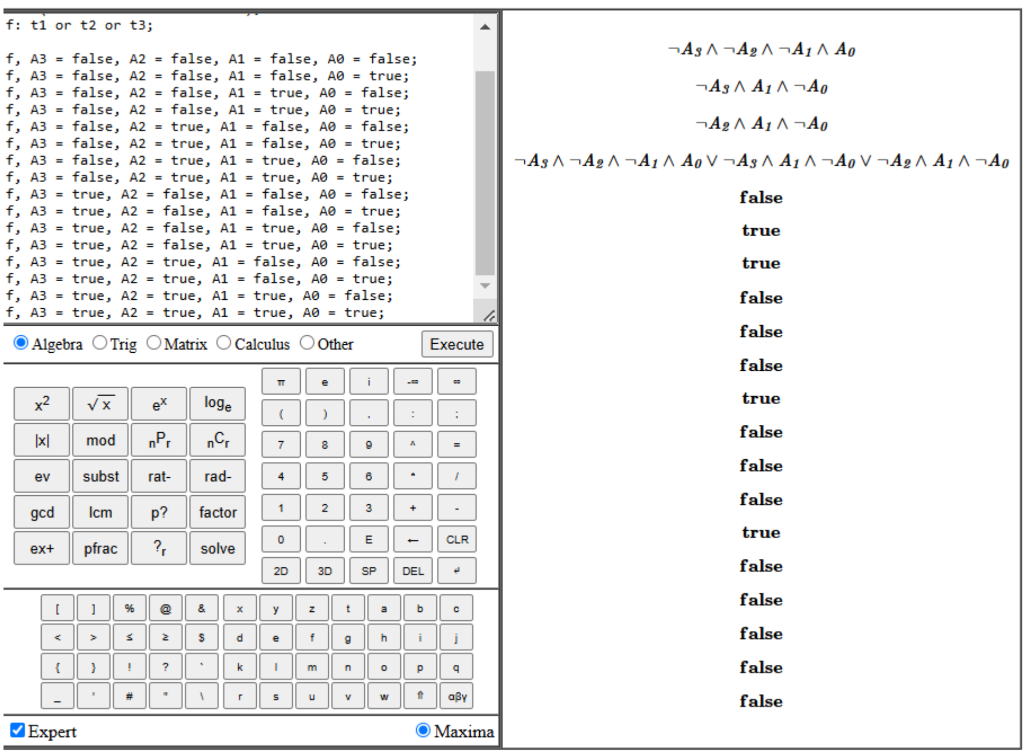
		A1 A0			
		00	01	11	10
A3 A2	00	0	0	1	0
	01	0	1	1	0
	11	0	0	0	0
	10	0	1	0	0

Equation: O3 = A3'A1A0 + A3'A2A0 + A3A2'A1'A0. (minimum SOP)

Equation Testing:

Testing for O0:

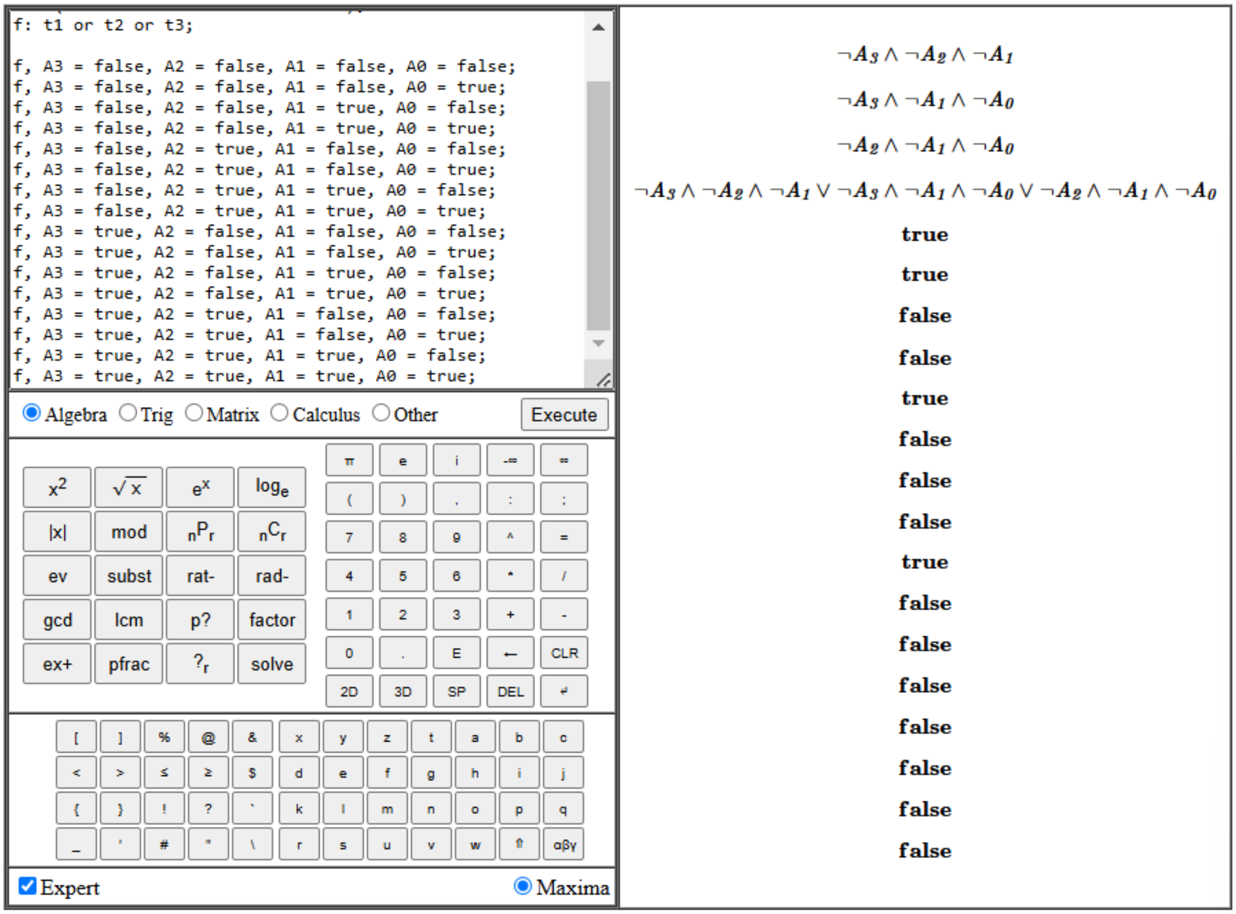
```
t1: (not A3 and not A2 and not A1 and A0);
t2: (not A3 and A1 and not A0);
t3: (not A2 and A1 and not A0);
f: t1 or t2 or t3;
```



The Maxima testing for O0 equations give true on 0001 (1), 0010 (2) ,0110 (6), 1010 (10) and false on the rest, as expected in the truth table above.

Testing for O1:

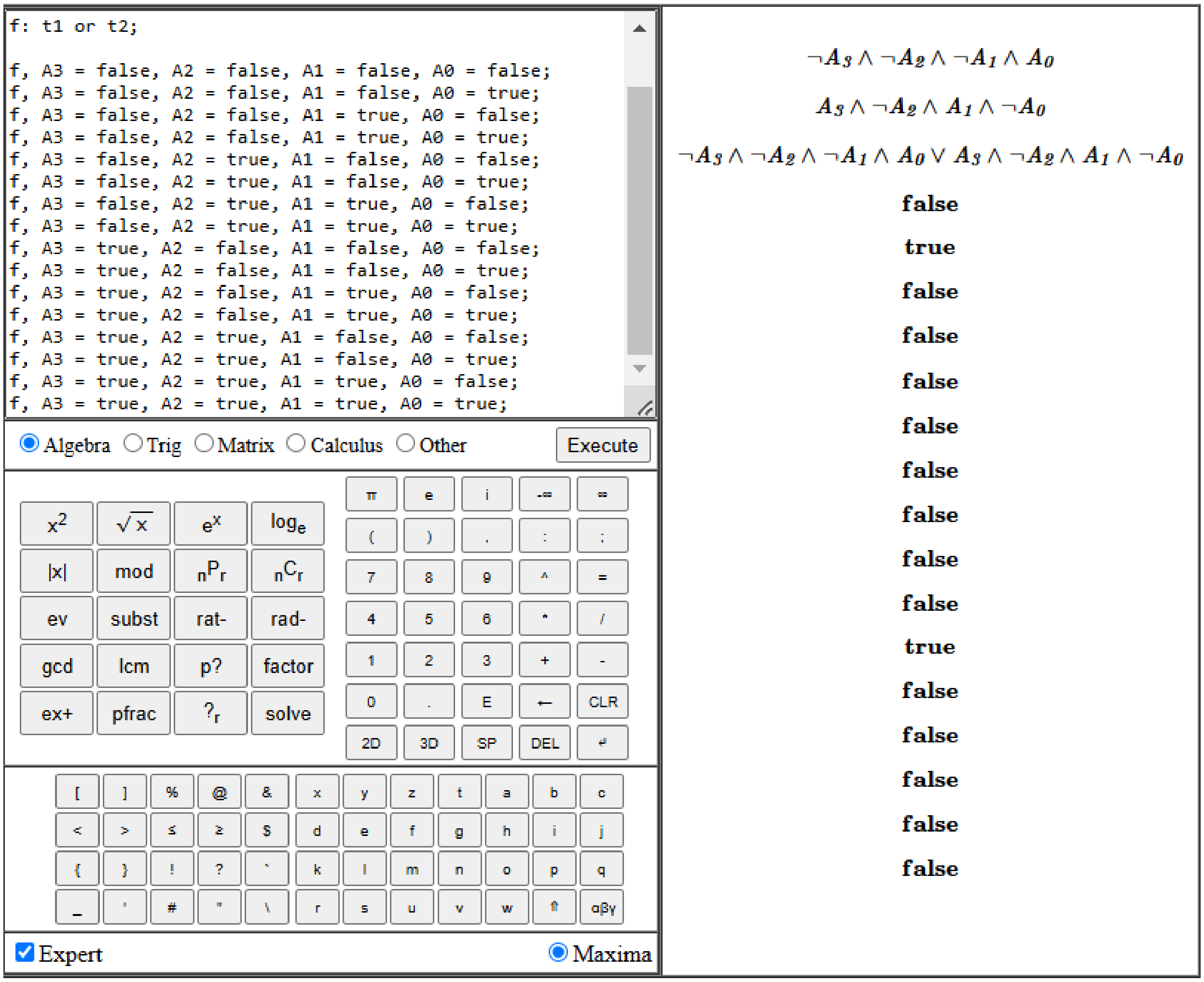
```
t1: (not A3 and not A2 and not A1);
t2: (not A3 and not A1 and not A0);
t3: (not A2 and not A1 and not A0);
f: t1 or t2 or t3;
```



The Maxima testing for O1 equations give true on 0000 (0), 0001 (1), 0100 (4), 1000 (8) and false on the rest, as expected in the truth table above.

Testing for O2:

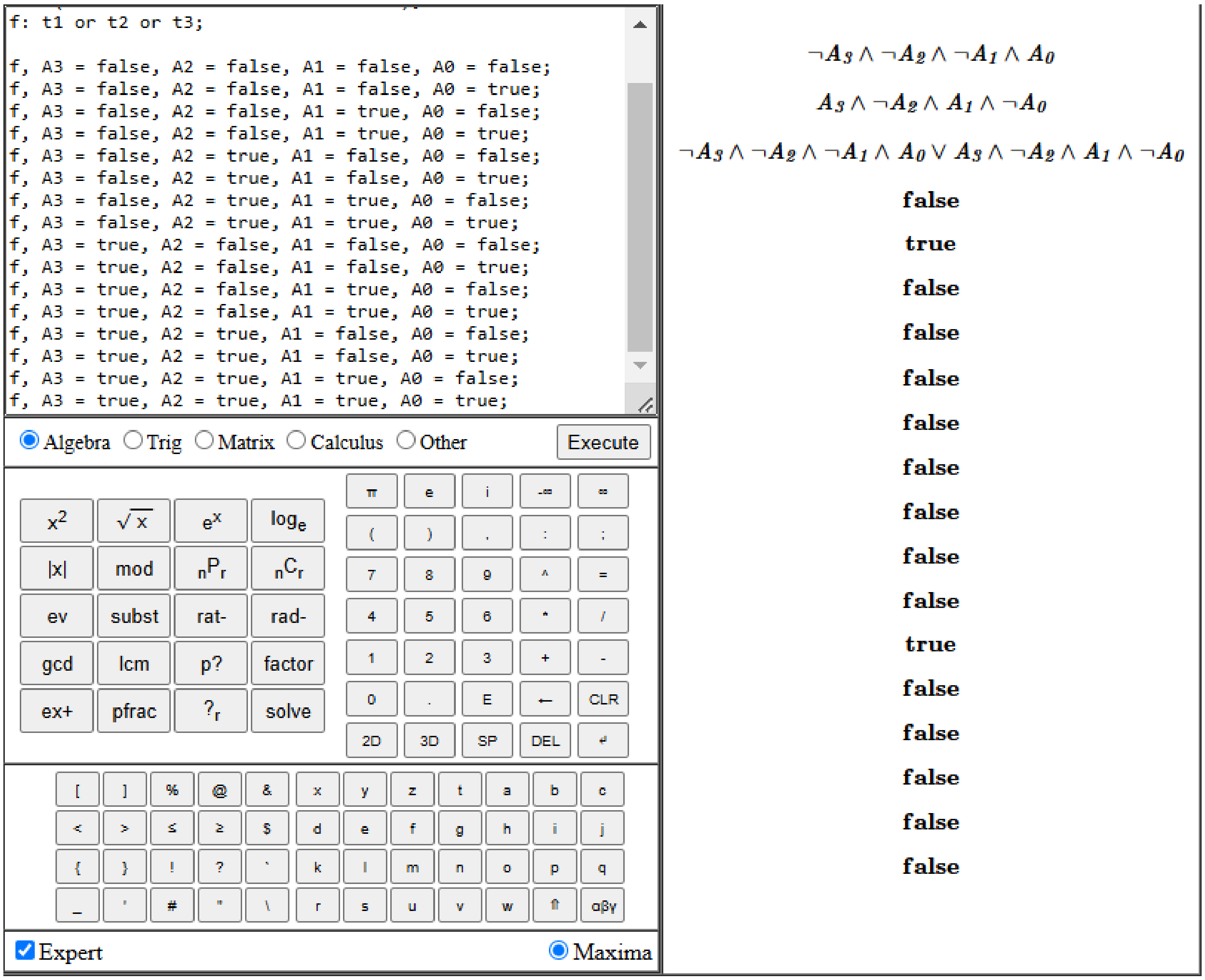
```
t1: (not A3 and not A2 and not A1 and A0);
t2: (A3 and not A2 and A1 and not A0);

f: t1 or t2;
```



The Maxima testing for O2 equations give true on 0001 (1), 1010 (10) and false on the rest, as expected in the truth table above.

Testing for O3:

```
t1: (not A3 and A1 and A0);
t2: (not A3 and A2 and A0);
t3: (A3 and not A2 and not A1 and A0);
f: t1 or t2 or t3;
```



The Maxima testing for O2 equations give true on 0011 (3), 0101 (5), 0111 (7), 1001 (9) and false on the rest, as expected in the truth table above.

Summary:

Since we had 4 different bits for output on every input, we tested their respective equations on maxima that worked as expected. O3 to O0 combined will show the outputs for every combination 4 bit input A3 to A0.

CP 220 - Project Phase I (Not Revised)

Project Topic: Digits of e

Student Name: Moosa Ahmed

Student ID: 169046796

Description:

This project focuses on creating binary representation of decimal digits from the mathematical constant *e*, values approximately **2.718 281 828 5**. The system takes a 4-bit binary input representing a decimal place (ranging from 0 to 10) and returns a 4-bit binary output corresponding to the digit of *e* at that position. For an input of zero, the system outputs the binary representation of '2' (equals '0010' in binary) which is the integer part of *e*.

Input:

The system accepts a 4-bit binary input A0, A1, A2 and A3. The value ranges from 0000 (0) to 1010 (10), showing the decimal place like 1010 (10) shows the 10^{th} decimal place of e that is number 5. The most significant bit will be A3 while A0 being the least significant. The input acts as an address, guiding the system to fetch the corresponding digit.

Output:

The system provides a 4-bit binary output from O0, O1, O2 and O3 with O3 being the most significant bit while O0 being the least significant bit, representing the decimal digit of *e* at the given position. For example, if the input corresponds to the third decimal place, the output will be 8 which will be represented as 1000 in binary. When the input is 0000 (0), the system outputs '2' (0010 in binary), showing the integer part of e.

Some sample inputs and their corresponding outputs:

'e' to 10 decimal places = 2.718 281 828 5

Input - binary	Decimal place (position)	Value fetched	Output - binary
0000	0^{th} decimal place (integer part)	2	0010
0011	3^{rd} decimal place	8	1000
0110	6^{th} decimal place	1	0001
1000	8^{th} decimal place	2	0010
1010	10^{th} decimal place	5	0101

Ambiguity:

No such ambiguity should occur, all inputs will give their corresponding decimal place value.

Error conditions:

Any value which is bigger than 1010 (10) will be considered invalid input which will give the output as 0000 (0).

NOTE: there is no '0' in the e value to 10 decimal places = 2.718 281 828 5

This will make sure the input was invalid (out of range) so output is 0000 (0).